

Elementary Number Theory (EC) Syllabus

Spring semester 2026

Course Title	Elementary Number Theory		Course Code	MC3112	Credit	3	Classroom	형설관 407
			Type	Advanced	Hours	4		
Teacher	Name	Class no.	Office	Office hours*	Contact	E-mail		
	임종렬	EC1	형설관 508	Tue3, Wed3, Fri3	051-606-2204	limjongryul@ksa.hs.kr		

***Office Hours:** A dedicated time for students to freely ask questions and engage in discussions with the instructor about lecture or lab materials they need help understanding or wish to explore further.

1. Course Description

Number Theory is known as the “Queen of Mathematics,” as it is one of the most beautiful areas in all of mathematics. The subject is famous for many elegant problems which are simple to state. Some of these have simple solutions which have been known for thousands of years, while others have frustrated the attempts of the best thinkers for generations.

This course covers most of the results in elementary number theory as listed below. Primary goal of this course will be introducing and proving the law of quadratic reciprocity, due to Gauss. Later on, we will talk about more advanced topics in number theory.

2. Core Target Competencies

Subject Core Competencies					
Problem solving	Reasoning	Communication	Attitudes and practices	Problem discovery and definition	Information ethics
Solving problems related to fundamental topics of mathematics, Proving and disproving statements; Using correct rules of logic, Write mathematical reasonings in a clear and rigorous way, Learning course material in an organized way, Modelisation of concrete problems using mathematics, Use critical thinking to analyze claims.					

3. Textbooks and References

A. Textbook : D. Burton, Elementary Number Theory 7th Edition

B. Reference

- J. H. Silverman, A friendly introduction to number theory
- G. H. Hardy and E. M. Wright, An Introduction to the Theory of Numbers

4. Class Method

- A. Lecture on new concepts
- B. Practice problem sessions
- C. Homework and quizzes

5. Evaluation

A. Evaluation Criteria

Activities	Midterm Exam	Final Exam	Attendance	Sum
25%	30%	35%	10%	100%
50 pts	60 pts	70 pts	20 pts	200 pts

B. Evaluation Method

(1) Activities

- Homework
- Quizzes
- Mini-research

(2) Exams

- Mid-term Exam: after studying section 6.3 / 3 hrs
- Final Exam: after studying Partition Theory / 3 hrs

(3) Attendance

- absence: -1.5 points except illness or official cases
- late : -0.5 point

6. Criteria for Awarding Credits

👉 Students with excellent academic achievement are recommended for the Academic Excellence Award and the Principal's Award, and are recorded according to core competencies in the Subject Details. However, students with poor academic attitude or poor attendance are excluded.

- Attendance for at least three-quarters of the total class hours is required to earn credit.
- Calculation of the basis score for awarding credit to absentees : In case of an examination missed due to official leave or medical leave, the score will be calculated by multiplying (the score corresponding to the total percentile score earned by the student on other examinations in the same subject) by (100% for official leave, or 80% for medical leave).
- If a basis score cannot be calculated, the grading method : A make-up examination will be conducted.

※ Refer to the "Academic Performance Management Regulations" for the percentage of credit points applied.

7. Weekly Plans

주(기간)	주제 및 내용	(비고)
제1주 2.23.~3.1.	1.1 Mathematical Induction 1.2 The Binomial Theorem	3.2.(월) 대체휴일
핵심역량		
제2주 3.2.~3.8.	2.2 The Division Algorithm 2.3 The Greatest Common Divisor 2.4 The Euclidean Algorithm	
핵심역량		
제3주 3.9.~3.15.	2.5 The Diophantine Equation 3.1 The Fundamental Theorem of Arithmetic 3.2 The Sieve of Eratosthenes	
핵심역량		
제4주 3.16.~3.22.	3.3 The Goldbach Conjecture 4.2 Basic Properties of Congruence 4.4 Linear Congruences and the Chinese Remainder Theorem	
핵심역량		
제5주 3.23.~3.29.	5.2 Fermat's Little Theorem and Pseudoprimes 5.3 Wilson's Theorem	
핵심역량		
제6주 3.30.~4.5.	6.1 The Sum and Number of Divisors 6.2 The Möbius Inversion Formula 6.3 The Greatest Integer Function	
핵심역량		
제7주 4.6.~4.12.	Review Week	
핵심역량		
제8주 4.13.~4.19.	중간고사 (4.14~4.17.)	4.13.(월) 정상수업

주(기간)	주제 및 내용	(비고)
제9주 4.20.~4.26.	7.2 Euler's Phi-Function 7.3 Euler's Theorem 7.4 Some Properties of the Phi-Function	
핵심역량		
제10주 4.27.~5.3.	8.1 The Order of an Integer Modulo n 8.2 Primitive Roots for Primes	
핵심역량		
제11주 5.4.~5.10.	9.1 Euler's Criterion 9.2 The Legendre Symbol and Its Properties 9.3 Quadratic Reciprocity	5.5.(화) 어린이날
핵심역량		
제12주 5.11.~5.17.	9.4 Quadratic Congruences with Composite Moduli 11.2 Perfect Numbers 12.1 The Equation $x^2 + y^2 = z^2$	
핵심역량		
제13주 5.18.~5.24.	12.2 Fermat's Last Theorem 13.2 Sums of Two Squares	
핵심역량		
제14주 5.25.~5.31.	13.3 Sums of More Than Two Squares Partition Theory	5.25.(월) 대체공휴일
핵심역량		
제15주 6.1.~6.7.	Review week	6.3.(화) 지방선거일
핵심역량		
제16주 6.8.~6.14.	보강 주간	
제17주 6.15.~6.19.	기말고사 (6.18~6.19.)	

※ Note: Lecture plans may be adjusted based on changes to the academic schedule.

8. Problem list

Section	Problem Number
1.1	1de, 4, 7, 8, 10, 14
1.2	1a, 3, 5, 8, 9, 10
2.2	4, 5, 7, 11
2.3	4e, 6c, 12, 13a, 14b, 16, 18, 20, 23
2.4	1, 2, 3, 5b, 6, 10c, 11, 12
2.5	2, 5, 6, 7, 8
3.1	3cd, 6a, 9b, 13, 14, 16a
3.2	4bc, 7, 8, 9, 10, 11, 12c
3.3	6, 7, 8, 9, 13, 16, 20, 25, 26b, 27, 28a
4.2	3, 4, 5, 6, 10, 12ab, 13, 17, 18
4.4	1df, 2, 4, 6, 7, 8, 9, 10, 11, 15, 17, 18, 20
5.2	2, 10, 11, 12, 13, 14, 15, 20
5.3	1, 2, 3, 4, 5, 9, 10, 12, 14, 16, 17
6.1	3, 6, 7, 8, 10, 11, 13, 16, 20, 21, 22, 23
6.2	2, 3, 4, 5, 6, 7, 8
6.3	3, 6, 7, 10, 12, 13
7.2	1, 4, 7, 12, 13, 14, 16, 17, 18, 19
7.3	2, 5, 11, 12
7.4	1, 3, 5, 6, 7, 11, 12, 13, 15, 16
8.1	2c, 6, 7, 10
8.2	6, 7, 10, 12
9.1	1, 2, 3a, 5, 6, 7, 12, 13
9.2	1bd, 2bd, 6a, 7, 12, 13, 14, 15, 18
9.3	1e, 3, 4, 5, 7, 9, 10, 11, 14, 16, 18, 19, 20
9.4	2, 5, 8, 10
11.2	3, 4, 7, 9, 11, 14, 15, 18, 20
12.1	ALL
12.2	ALL
13.2	ALL
13.3	ALL